JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 93, NO. B10, PAGES 11,721–11,733, OCTOBER 10, 1988

Mid-Cretaceous Paleomagnetic Reference Field for North America: Restudy of 100 Ma Intrusive Rocks From Arkansas

BRIAN R. GLOBERMAN[1] AND E. IRVING

Pacific Geoscience Centre, Geological Survey of Canada, Sidney, British Columbia

We have obtained new paleomagnetic data from three middle Cretaceous alkalic intrusive complexes in central Arkansas: Magnet Cove and Potash Sulfur Springs intrusions (about 100 Ma, 20 sites), and Granite Mountain stock (about 88 Ma, 6 sites). Out of 254 specimens measured, 147 from 20 sites were accepted. The remainders have scattered magnetization directions which experiments indicate were caused by lightning occurring during frequent electrical storms in this region. The geologic evidence indicates that the intrusions have not been tilted since emplacement. The directions of magnetization calculated with respect to present-day horizontal yield a mean paleopole at 192.5°E, 74.1°N ($N = 20$, $K = 34$, $A_{95} = 5.7°$). This does not differ at 95% confidence from that determined previously by Scharon and Hsu, but the error is diminished by half. The angular dispersion (S_d) of site virtual geomagnetic poles (VGPs) is 13.6°, similar to the value of S_d (about 15°) predicted by current paleosecular variation models. The revised Arkansas paleopole is combined with four other cratonic North American paleopoles of mid-Cretaceous age to yield a revised middle Cretaceous reference pole at 196°E, 71°N, $A_{95} = 4.9°$. The mean of nine paleopoles derived from overprints in rock units from the craton and craton margin, which range in age from Proterozoic to Jura-Cretaceous, is 202°E, 68°N, $A_{95} = 8.2°$. This is not different from the reference paleopole, indicating that overprinting was very widespread across North America in mid-Cretaceous time.

INTRODUCTION

Our purpose is to reconsider the mid-Cretaceous paleomagnetic reference field for North America. Such reference fields, expressed as paleopoles for cratonic blocks, fix the block relative to the Earth's axis of rotation and hence are essential in the study of the kinematics of accreted terranes. The study of the Cretaceous reference field began with the work of *Runcorn* [1956] based on three oriented samples from the Dakota Sandstone of Arizona, which yielded a paleopole in the Bering Sea (Table 1, entry A). During the early 1960s, more data became available, and it became possible to calculate mean paleopoles determined as an average of several rock units within a range of ages centered about the mid-Cretaceous. Notable additions were data from the Sierra Nevada batholiths [*Grommé and Merrill*, 1965; *Grommé et al.*, 1967] which were approximately concordant with data obtained from Mount Megantic [*Larochelle*, 1961] and the Monteregian Hills [*Larochelle*, 1962] in southeastern Quebec (Table 1, entry B), showing that the Sierra Nevada had not moved by any large amount relative to the craton nor had it undergone appreciable tilting. In the later 1970s the Sierra Nevada data fell into disuse because the possibility that small motions of the intrusions relative to the craton could not be ruled out. However, data of *Frei et al.* [1984] indicate that rotations and translations in the Sierra Nevada are still within the uncertainty limits. Hence the recent tendency has been to limit the reference data to those derived from localities firmly rooted in the craton and far distant from the Cordillera.

Gordon et al. [1984] calculated a best fit paleomagnetic Euler pole (PEP) for 13 cratonic paleopoles of Jurassic through Cretaceous age and generated a set of reference paleopoles for each 10-m.y. interval. Their paleopole for mid-Cretaceous time (~100 m.y.) is given in Table 1, entry F. A remarkable feature of Table 1 is that despite the accumulation of new data and the variation in analytical procedures, the mean reference paleopole for the mid-Cretaceous of North America has remained little changed. Why then is a reinvestigation necessary?

There are three reasons. First, studies of mid-Cretaceous rocks in the western North American Cordilleran and in the Arctic regions are now very numerous, and it is becoming increasingly desirable to know the reference paleopole as accurately and reliably as possible in order to make accurate tectonic reconstructions. Second, the mid-Cretaceous is becoming recognized as an interval during which widespread remagnetization of older rock units commonly occurred. Therefore a precisely determined reference field is required for accurate identification of magnetic overprints. Third, it is of great importance to determine the magnitude of the paleosecular variation (PSV) during the mid-Cretaceous, when reversals of the geomagnetic field were infrequent. We have restudied the paleomagnetism of the mid-Cretaceous Arkansas alkalic intrusions first described by *Scharon and Hsu* [1969]. In their work, about 50% of the data were rejected because they had scattered directions, for which no physical explanation was given at the time. Also the original data yield an angular dispersion far greater than would be anticipated from paleosecular variation models. In this paper we shall seek to explain this large dispersion and high rejection rate as being the result of lightning strikes.

Allan V. Cox, to whom this volume is dedicated, made considerable contributions to PSV analysis and to studies of the effects of lightning on the magnetism of rocks. Because we shall lean heavily on the ideas and techniques relevant to these two problems that he helped to develop, it is appropriate to describe his contributions to them.

In the early 1950s Fisher statistics [*Fisher*, 1953] became universally applied in paleomagnetic studies. Fisher statistics provide estimates of the mean, error, and dispersion of direc-

[1]Also at Earth Sciences Board, University of California, Santa Cruz.

Published 1988 by the American Geophysical Union.

Paper number 7B3036.

TABLE 1. History of the Determination of the Mid-Cretaceous (~100 Ma) Reference Field for Cratonic North America

	Study	N	Longitude, °E	Latitude, °N	A_{95}
A	*Runcorn* [1956]	1*	200	75	···
B	*Irving* [1964]	3	183	69	7
C	*McElhinny* [1973]	9	187	64	6
D	*Mankinen* [1978]	8	186	68	2
E	*Irving* [1979]	4	188	69	4
F	*Gordon et al.* [1984]	13†	190	70	···
G	this study	4	196	71	5

N, number of paleopoles used in determination of reference field; A_{95}, radius of 95% confidence circle for paleopoles.

*This study comprises only one rock unit (Dakota sandstone).

†This reference pole was determined using a mathematical best fit paleomagnetic Euler pole derived from 13 Jurassic and Cretaceous paleopoles; reference poles in the outer studies (except A) derived from *Fisher* [1953] average of N paleopoles.

tions of magnetization. Paleomagnetic observations obtained at localities spaced through a rock sequence sample the paleofield at irregular intervals. By obtaining several rock specimens at each site, experimental errors can be minimized, so that the dispersion of site-mean directions should provide an estimate of the magnitude of the PSV. These estimates, made by Fisher's method, could be compared with the dispersion of the recent geomagnetic field and with estimates of dispersion from other rock-units. In order to make these comparisons, models of the statistical behavior of the present and past geomagnetic field are required.

Creer [1955] provided a global analysis of directions of the recent geomagnetic field and subsequently [*Creer*, 1962], compiled the directions observed about a line of latitude, averaged them, and obtained their dispersion as a function of latitude. The dispersions calculated were the sum of the variations in the nondipole field and the effect of the equatorial dipole component, i.e., the inclination of the geomagnetic axis.

Analysis of field directions for PSV analysis has largely been superseded by analysis of virtual geomagnetic poles (VGPs). A VGP corresponds to the direction of magnetization calculated for a single observation of the paleofield. The dispersion of VGPs rather than site-mean directions may be used for estimating PSV, the assumption being that the dispersions are caused by variation in inclination of the geomagnetic axis only, i.e., by dipole wobble. This procedure was suggested independently by *Creer et al.* [1959] and by *Cox* [1959]. In a series of masterly papers, *Cox* [1962, 1964, 1970] then went on to model the effects of dipole wobble and nondipole field variations and compared them with his own and others' observations of the paleosecular variation, bringing PSV analysis into the state we know it today.

Another of Cox's contributions was to the study of the magnetic effect of lightning strikes. When lightning strikes, the current passes along the surface carrying with it a transient magnetic field, which induces a strong magnetization termed "isothermal remanent magnetization" (IRM). Such effects were bothersome to early mountaineers in Europe because they deflected magnetic compasses, and it is not surprising therefore that they became the first manifestation of IRM to be studied. With the resurgence of interest in the remanent magnetism of rocks in the 1950s, such effects were discovered not only in mountainous areas but also in level terrain where the incidence of electrical storms is high. Standard papers on this problem were written by *Graham* [1961] for the Pilansberg dykes in South Africa and by *Cox* [1961] for lava flows of Plio-Pleistocene age in Idaho's Snake River Plain. They both found that typical lightning strike, with a peak line current of about 22,000 A, would produce a cylindrical zone of anomalous magnetization with a radius of about 60 cm. They suggested that a minimum field of 0.5 mT is required to produce a lasting IRM. Curves for the strengths of lightning-induced fields attained at various distances from the discharge point of such a strike were given by *Cox* [1961, p. 155].

GEOLOGY AND SAMPLING

The Arkansas alkalic intrusions occur along the boundary between deformed Paleozoic rocks of the Ouachita Mountains to the northwest and undeformed Upper Cretaceous and Tertiary strata to the southeast [*Haley et al.*, 1976]. They crop out discontinuously along a 150-km zone from Murfreesboro (Prairie Creek intrusion, about 106 Ma) to Little Rock (Granite Mountain stock, about 88 Ma (Figure 1)). The intrusions sampled are Magnet Cove and Potash Sulfur Springs (MC/PS, Hot Spring and Garland counties) and Granite Mountain (GM, Pulaski County) south of Little Rock (Figure 1).

Generally, there is no direct evidence for or against tilting, as the igneous complexes are intruded usually into deformed Paleozoic rocks. However, vertical lamprophyric dikes of the Prairie Creek intrusion [*Morris*, 1987], dated at 106 Ma plus or minus 3 Ma [*Zartman*, 1977], cut flat-lying strata of the Lower Cretaceous Trinity Group [*Haley et al.*, 1976]. Hence we infer that the intrusions have probably not been tilted. The paleohorizontal required for paleomagnetic analysis is therefore the present horizontal plane. The Prairie Creek intrusion itself was not sampled because of poor exposure and extreme weathering.

Magnet Cove Complex

Magnet Cove is a ring-dike complex for which *Zartman et al.* [1967] obtained K-Ar biotite ages of 97 Ma from alkali syenite and 99 Ma from garnet ijolite. They also determined a Rb-Sr age of 101 Ma from the garnet ijolite. *Naeser and Faul* [1969] obtained a fission track apatite age of 97 Ma for a carbonatite and a fission track sphene age of 102 Ma for a nepheline syenite. *Eby* [1987] reported an average fission track sphene age for the complex at 101.4 plus or minus 1.0 Ma. Fission track apatite ages determined for all of the other intrusive units fall between 95.9 and 101.4 Ma. Where fission track apatite and sphene ages were determined for the same sample, the apatite ages were always younger, suggesting that there were two periods of intrusion. The silicate units were emplaced during the first event, the carbonates during the second [*Eby*, 1987].

According to *Erickson and Blade* [1963] the Magnet Cove complex is zoned concentrically. Its core consists mainly of carbonatite. Its rings comprise trachytes and syenites. Masses of magnetite-rich pyroxenite (jacupirangite), the most mafic rock unit of the suite, occur along the northeastern margin of the complex. Small leucocratic dikes cut all major units. Based on these field relations, *Erickson and Blade* [1963] proposed the following age sequence, from oldest to youngest: (1) phonolite and trachyte, (2) jacupirangite, (3) alkalic syenites, (4) ijolite, and (5) carbonatite and minor dike rocks. Pet-

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

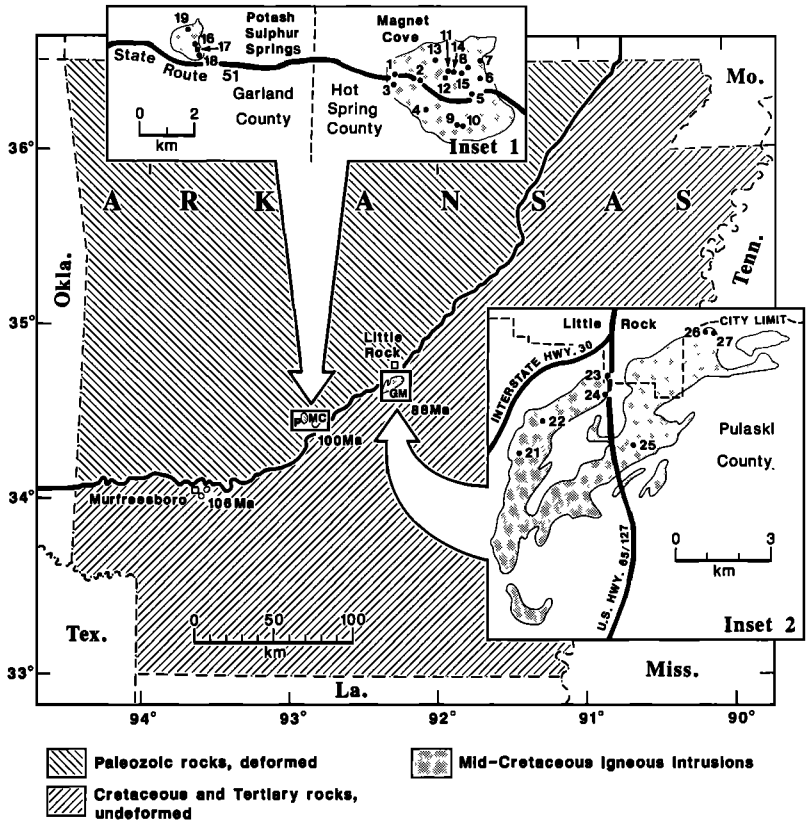


Fig. 1. Location of 100 Ma Magnet Cove and Potash Sulfur Springs (MC/PS), and 88 Ma Granite Mountain (GM) igneous intrusive complexes. Sampling sites at MC/PS and GM are shown in insets 1 and 2, respectively. Lamprophyric dikes of the 106 Ma Prairie Creek intrusion (not sampled) crop out near Murfreesboro in southwestern Arkansas. Geology after *Haley et al.* [1976].

rological and geochronological evidence therefore indicates that the various rock units were intruded at different times, not simultaneously. Each of these rock units has been sampled, and each probably represents an independent interval of time.

Between five and nine individually oriented cores were collected from 15 sites. All magnetic orientations were checked by sun compass. Compass anomalies of up to 45° are observed, jacupirangite giving the largest anomalies, as noted previously [*Williams*, 1891; *Erickson and Blade*, 1963]. The average intensity of natural remanent magnetization (NRM) ranges from 0.02 to 8.8 A m^{-1} (Table 2).

Potash Sulfur Springs Complex

The Potash Sulfur Springs complex is dated at 100 ± 2 Ma on the basis of $^{207}Pb/^{206}Pb$ zircon ages from feldspathoidal syenite [*Zartman and Howard, 1987*]. *Eby* [1987] reported fission track apatite ages of 99.8 Ma from ijolite and 102.4 Ma from carbonatite. Like the Magnet Cove complex, it is circular in shape although smaller (1.5 km) in diameter [*Howard*, 1974, D. W. Pollack, unpublished map, 1966]. According to *Howard* [1986] the Potash Sulfur Springs complex is a nepheline syenite stock intruded by mafic differentiates (e.g., jacupirangite and ijolite) and carbonatite. Major element analyses of micas, pyroxenes and amphiboles indicate four generations of carbonatite intrusion and/or metasomatism [*McCormick*, 1986], although geochronologically these events are indistinguishable [*Eby*, 1987]. Hence unlike Magnet Cove, there is no firm evidence that each of the sites record a discrete time of cooling, although they probably do. Four separate sites were sampled in small creeks at UMETCO vanadium quarry (Figure 1, inset 1.)

Granite Mountain Stock

The Granite Mountain stock has yielded ages ranging from 91 to 86 Ma, based on K-Ar biotite and Rb-Sr dating [*Zartman et al.*, 1967], and on fission track dating of apatite and sphene [*Morris and Eby*, 1986; *Eby*, 1987]. The Granite Mountain stock is the most leucocratic of the Arkansas intrusions [*Morris and Eby*, 1986] and, unlike the Magnet Cove and Potash Sulfur Springs complexes, is not distinctly zoned [*Williams*, 1891]. Instead, it is a complex mixture of olivine syenite and three distinct metasomatic phases: Pulaskite, a nepheline syenite produced during the first phase, is the dominant rock unit [*Morris*, 1987]. Seven sites at four localities in roadcuts and quarries were sampled (Figure 1, inset 2).

LABORATORY PROCEDURE

A total of 254 specimens were cut from 172 cores collected at 26 sites. Measurements were made on a modified digital spinner magnetometer controlled by a microcomputer. The system used provides four independent measurements of the magnetization vector, along with the precision (k) of each determination. The latter quantity ranges from 10^2 to 10^3 for homogeneous specimens in which the internal alignment of magnetization directions of individual particles is good, to below 5, when the internal alignment approaches a random distribution and the data become meaningless.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Site-Mean Directions and Virtual Geomagnetic Poles for Alkalic Intrusions, Arkansas

Site	Rock Type	n/n'	NRM	Direction		k	Virtual Geomagnetic Pole			
				Dec, deg	Inc, deg		α_{95}	Longitude °E	Latitutde °N	A_{95}
MC-01	SNS	5/5	2.5	352	61	91	8	227	80	11
MC-02	TPH	6/6	4.7	360	68	12	20	266	73	32
MC-03	GPS	6/6	0.8	335	55	13	19	186	69	22
MC-04	GPS	5/5	3.7	330	42	80	9	164	62	8
MC-05*	TPH	5/5	4.6	310	50	10	25	187	48	27
MC-06*	APB	6/6	10.6	307	80	6	30	246	44	79
MC-07	JAC	9/9	87.5	333	60	65	7	202	67	8
MC-08	JAC	8/9	14.1	355	60	176	4	236	83	5
MC-09	GPS	11/11	2.6	324	53	56	6	186	60	7
MC-10	NSP	9/9	0.2	354	55	43	8	191	85	9
MC-11	APB	12/12	2.4	341	58	103	4	198	75	5
MC-12	TIN	10/11	0.4	333	50	39	8	174	67	8
MC-13	GPS	5/5	0.8	329	57	26	15	192	65	18
MC-14	GPS	7/9	2.5	315	71	12	19	226	53	33
MC-15	JAC	5/5	6.9	355	56	154	6	205	85	7
PS-16	IJO	8/8	1.1	337	54	46	8	182	71	9
PS-17	SYE	7/7	1.9	334	55	64	8	188	69	9
PS-18*	IJO	2/2	1.0	336	58	22	56	197	70	72
PS-19*	CAR	4/4	1.7	319	52	11	29	186	56	32
GM-20	PUL	9/9	0.8	333	45	40	8	166	65	8
GM-21	PUL	4/4	0.7	347	59	587	4	207	79	5
GM-22	PUL	5/5	2.9	336	45	41	12	163	68	12
GM-23*	PUL	5/5	1.1	300	59	11	25	203	43	32
GM-24	PUL	7/8	0.8	021	53	12	19	355	73	21
GM-25	PUL	9/10	4.8	335	49	30	10	171	69	10
GM-26*	PUL	5/5	5.0	347	65	5	38	233	75	61

n/n', number of specimens used in determination of site-mean direction, n' total number of specimens collected; NRM, average intensity of natural remanent magnetization of site, in 10^{-1} A m^{-1}; Dec, declination; Inc, inclination; k, (K) *Fisher* [1953] precision parameter for directions (VPGs); α_{95} (A_{95}), radius of 95% confidence circle for directions (VGPs); MC, Magnet Cove; PS, Potash Sulfur Springs; GM, Granite Mountain; SNS, sphene nepheline syenite; TPH, trachyte phonolite; GPS, garnet pseudoleucite syenite; APB, altered phonolite and breccia; JAC, jacupirangite; NSP, nepheline syenite pegmatite; TIN, tinguaite dike; IJO, ijolite; SYE, nepheline syenite; CAR, carbonatite; PUL, pulaskite. Lithologies from *Erickson and Blade* [1956, 1963].

*Sites omitted from calculations.

All directions of natural remanent magnetization (NRM) are plotted in Figure 2*a*. About half of the specimens show a strong clustering directions (the "main cluster") to the west of north, with downward inclinations of about 60°. Its center is displaced about 15° from the present Earth's field (PEF) direction. The rest of the specimens have an essentially random distribution and are termed "aberrant."

Eighteen aberrant specimens were thermally demagnetized in a stepwise manner to a temperature of 600°C. Most had high k values, but none migrated into the main cluster; they did not respond to thermal treatment. The remaining 228 specimens were demagnetized in alternating fields (af). Of these, 182 specimens (minimum of three per site) were demagnetized using four or more steps, and the principal component of magnetization was determined based on the results of the computer program LINEFIND [*Kent et al.*, 1983]. Remanent magnetizations of the remaining 46 specimens were partially demagnetized using three af steps, and the principal component was determined by vector subtraction. The latter technique was considered appropriate for those specimens with high k values, whose directions reached a stable endpoint similar to that of the "main cluster" by the second af step.

DESCRIPTION OF MAGNETIZATIONS

The magnetizations are of three general types. Type A have NRM directions that are well grouped, directed to the north or northwest, and comprise the central part of the main cluster of Figure 2*a*. The magnetizations are characterized by a relatively slow decrease in intensity during af demagnetization (i.e., high coercivity), stable endpoints, and k values in the range of 10^2 to 10^3 (Figure 3). Type A magnetizations are found in 23% of specimens.

Type B magnetizations are characterized by NRM directions that are greatly scattered. They comprise 38% of all specimens studied and about 50% of the aberrant group in Figure 2*a*. In these specimens there is a rapid decrease in intensity during af demagnetization which indicates low coercivity (Figure 4*b*). Nonetheless, by about 10 mT, an essentially stable endpoint direction is obtained (Figure 4). This stable remanence, which comprises only a small proportion of the NRM, has high coercivity. Values of k exceed 50 and, like the type A magnetization, the direction is downward to the north and northwest. Type B magnetizations are that part of the aberrant group which responds well to af (but not thermal) treatment.

Type C magnetizations (39% of specimens) are characterized either by the absence of a recognizable stable endpoint or by a stable remanent direction that is more than two angular standard deviations from the direction of the "main cluster." Figure 5 is an example of the latter. The NRM direction is close to the PEF, but the stable magnetization is directed to the south. Type C magnetizations are not systematically directed. Indeed, some directions are upward, indicating reverse polarity. Upward directions are most probably spurious

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

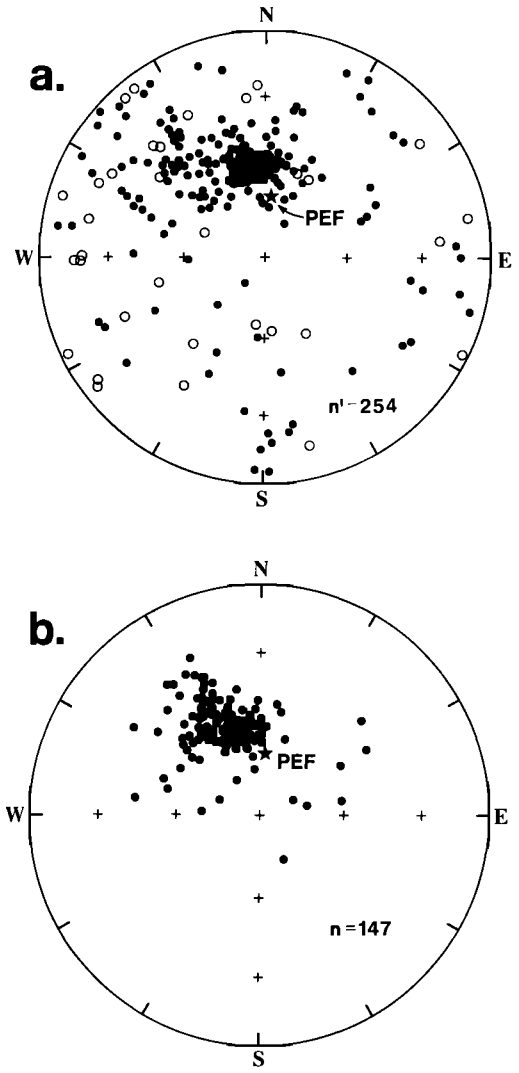


Fig. 2. Equal-area projections show (*a*) initial natural remanent magnetization (NRM) directions from $n' = 254$ specimens, and (*b*) the directions of stable magnetization of $n = 147$ specimens used in calculating site-mean directions for Arkansas intrusions. Perimeter is the present horizontal plane at the sampling sites. Note that both groups of directions are displaced from the present Earth's field direction (PEF).

because the bodies were intruded during the Cretaceous Long Normal Superchron (118–83 Ma [*Harland et al.*, 1982]). Type C magnetizations typically, but not always, show rapid decreases in intensity and within-specimen precision (k) during af demagnetization. Low values of k (less than 20) may partly account for the apparent absence of a stable endpoint in many specimens.

In no instance was thermal demagnetization effective in altering the aberrant yet stable components of type C magnetizations. The univectorial decay is shown in the orthogonal plots (Figure 6). Note also that the intensity decay is approximately linear, the unblocking temperatures being more-or-less uniformly distributed between room temperature and 600°C. This could have one of several causes. For example, the Curie temperatures could be widely distributed, or the magnetic material could be superparamagnetic. However, we suggest that the type C magnetization is an isothermal remanent magnetization (IRM) and that the most relevant comparison is with the decay characteristics of IRM acquired in fields of moderate strength below the saturating field, such as those observed by *Petrova* [1957, p. 64] and by *Kobayashi* [1959, p. 105]. The most common cause of IRM in nature is lightning [*Humboldt*, 1797; *Cox*, 1961; *Graham*, 1961]. The idea that lightning is responsible could explain the high intensities commonly observed (Table 2) and the highly scattered aberrant magnetizations (Figure 2*a*). It also provides a reasonable explanation of some type B magnetizations, i.e., a lightning-induced IRM, removable by af demagnetization, being superposed on a moderately stable thermoremanent magnetization (TRM). Moreover, electrical storms are frequent in the southeastern United States.

NATURAL AND SYNTHETIC IRM

To test the above idea, the following experiment was conducted. First, IRM acquisition curves were determined for typical specimens. The specimens reached saturation by 150–200 mT and have bulk coercivities ranging from 15 to 30 mT. This indicates that lightning-induced fields of this magnitude could produce a large IRM (Figure 7). Eleven cores (from 10 sites) with type B and C magnetizations were then selected.

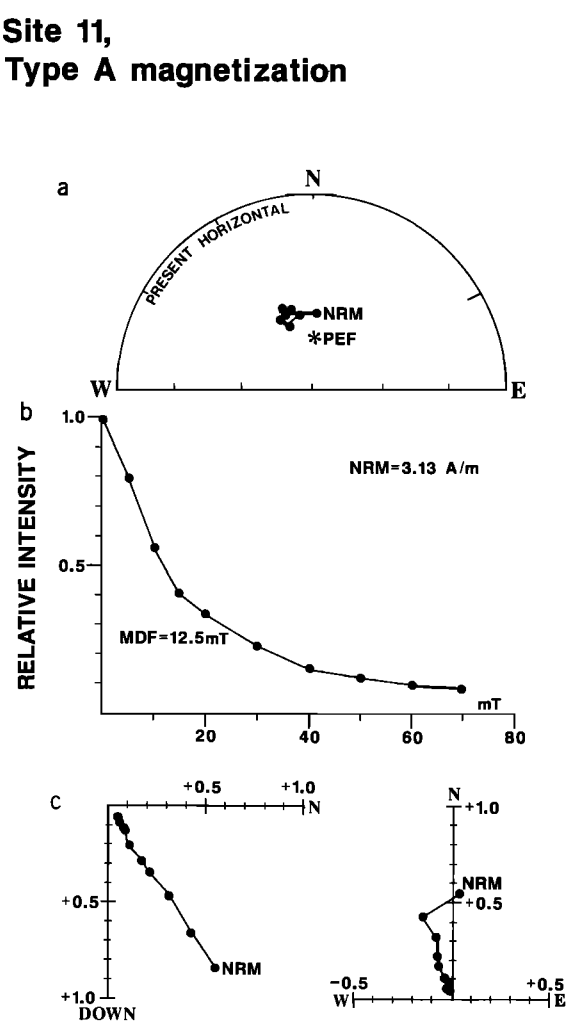


Fig. 3. Alternating field (af) demagnetization of type A magnetization at site 11, Magnet Cove intrusion. (*a*) Changes in direction, (*b*) changes in intensity, (*c*) orthogonal plots. Note the high coercivity (Figure 3*b*), and the stable endpoint, distinct from the PEF (Figure 3*a*), that is reached by 15 mT (Figure 3*c*).

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

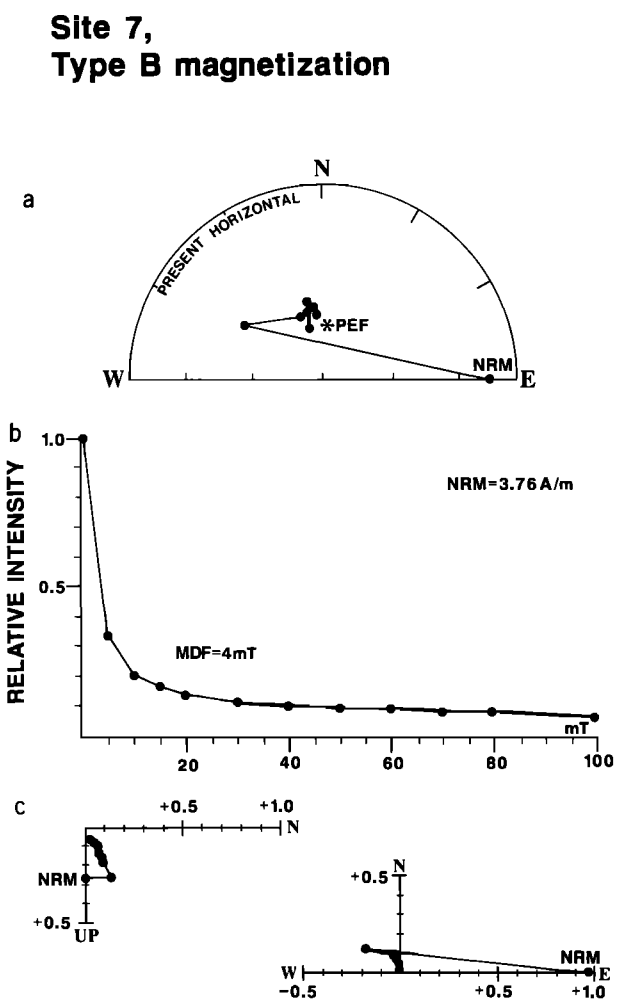


Fig. 4. The af demagnetization of type B magnetization at site 7, Magnet Cove intrusion. (*a*) Changes in direction, (*b*) changes in intensity, (*c*) orthogonal plots. Note the low coercivity (Figure 4*b*), aberrant NRM direction (Figure 4*a*) and the stable endpoint, distinct from the PEF, that is reached by 10 mT (Figure 4*c*).

Two specimens were cut from each core, and one specimen of each pair was thermally demagnetized (Figure 8*a*). Except at sites 5 and 8B, the intensity decrease occurring from room temperature to between 525° and 600° C is essentially linear.

Other specimens of each pair (eight were studied) were given IRMs by the following procedure. First, they were demagnetized in alternating fields of 100 mT to minimize the NRM. Six were then given IRMs by applying steady fields ranging from 2.5 to 50 mT in a solenoid. The remaining two specimens were given IRMs in fields of 100 and 450 mT in an electromagnet. Following this, they were af demagnetized at 0.5 mT in order to remove very soft viscous remanent magnetizations (VRMs). All specimens were then thermally demagnetized at temperatures of 50°–600°C in 50°C increments. The decay curves (labeled with the applied field strength in Figure 8*b*) varied from concave upward to convex upward. Decays were linear for IRMs acquired in fields between 5 and 50 mT. In the range 100 and 450 mT, as saturation is approached, the unblocking temperatures became more discrete, tending toward the square-shouldered form characteristic of TRM (see Figure 8*a*, specimen 8B).

The close similarity of the families of intensity decay curves for the NRMs and synthetic IRMs (Figure 8) suggests that the type B and C magnetizations observed in nine out of 11 specimens are actually lightning-induced IRMs which may have been acquired in fields ranging in strength from 5 to 50 mT. The intensity decay curves of the other two specimens (sites 5 and 8) resemble those of the IRM of specimens magnetized at very low (2.5 mT) and high (450 mT) fields, respectively. According to the table of *Cox* [1961, p. 155] these fields correspond to distances of about 10–50 cm from the lightning cur-

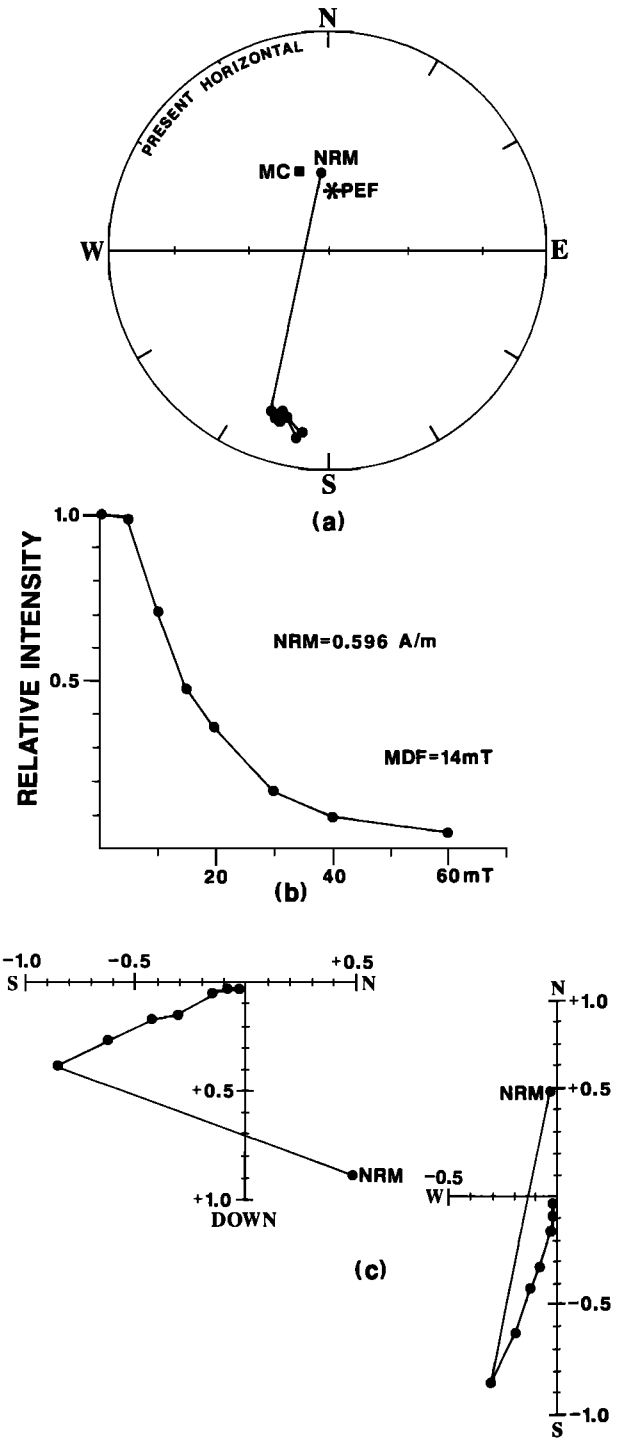


Fig. 5. The af demagnetization of type C magnetization at site 22, Granite Mountain stock. (*a*) Changes in direction, (*b*) changes in intensity, (*c*) orthogonal plots. Note the high coercivity (Figure 5*b*), and the nearly perfect univectorial decay in alternating fields between 5 and 40 mT (Figure 5*c*). The stable remanence is displaced about 100° (Figure 5*a*) from both the PEF and the "main cluster" (MC).

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

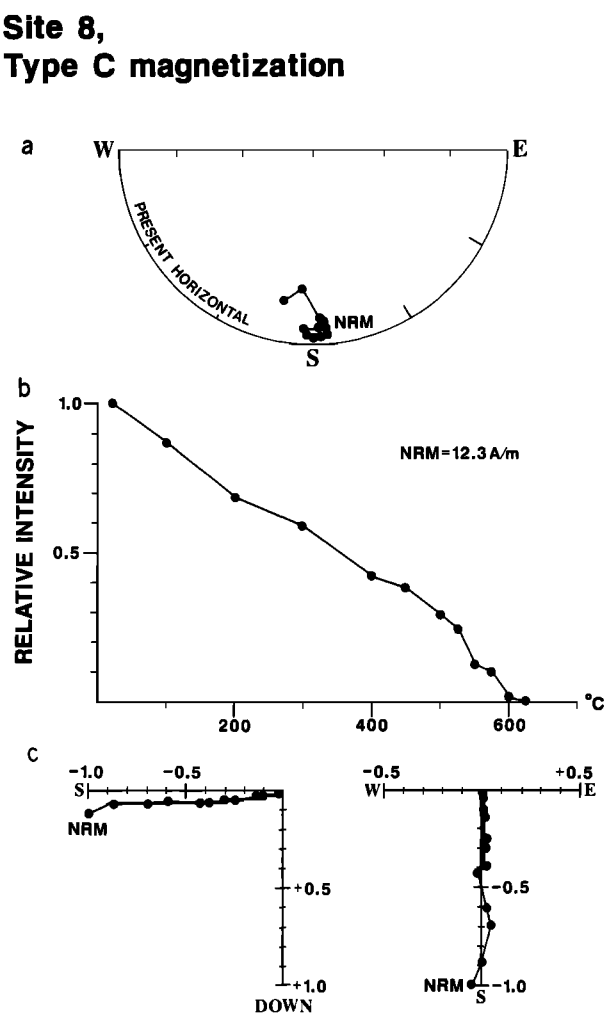


Fig. 6. Thermal demagnetization of type C magnetization at site 8, Magnet Cove intrusion. (*a*) Changes in direction, (*b*) changes in intensity, (*c*) orthogonal plots. Note linear intensity decay to 600°C (Figure 6*b*), and stability of the shallow, south-down direction from NRM to 580°C (Figures 6*a* and 6*c*).

rent itself. In a cylinder of 60-cm radius centered about the lightning current, about 80% of the volume lies between 13 and 50 cm from the center. Hence within the zone of anomalous magnetization surrounding the lightning current, about 80% of the rocks would have been subjected to peak fields ranging in strength from 5 to 50 mT [*Cox*, 1961]. Peak fields in excess of 50 mT should occur in only about 5% of the zone. These estimates compare well with the laboratory data of Figure 8. Hence we propose that the hard secondary magnetizations found in most "aberrant" specimens of the Arkansas intrusions are IRMs acquired in lightning-induced fields of mainly 5–50 mT, with a small proportion (two out of 11) in fields weaker than 5 mT. It is to be expected that the original TRM in some rocks was thoroughly obliterated by the IRM either because it was soft or because the lightning current was close at hand; such magnetizations are termed type C. We favor the latter explanation because the coercivity of selected specimens with type C magnetization is sometimes high (about 27 mT, see Figure 7). In rocks with type B magnetization the original TRM was preserved because the field produced by the lightning current was less than the coercivity of the TRM. This would arise either because the coercivity of the initial TRM was high or the lightning current was weak (i.e., the lightning strike was not close).

MEAN DIRECTION AND PALEOPOLE

A total of 80 specimens, most with type C magnetizations, were omitted from analysis for the above reasons. Site-mean directions and corresponding VGPs are listed in Table 2. Six sites comprising 27 specimens were omitted from further analysis because their errors (α) were greater than 20° and were considered too large. The remaining 147 specimens from 20 sites (out of a total of 172 cores collected at 26 sites) were used in calculating the mean direction and paleopole (Tables 2 and 3 and Figures 2*b* and 9*a*). This corresponds to a rejection rate of 42%, which is close to the 50% of *Scharon and Hsu* [1969].

Mean directions for three intrusions are plotted in Figure

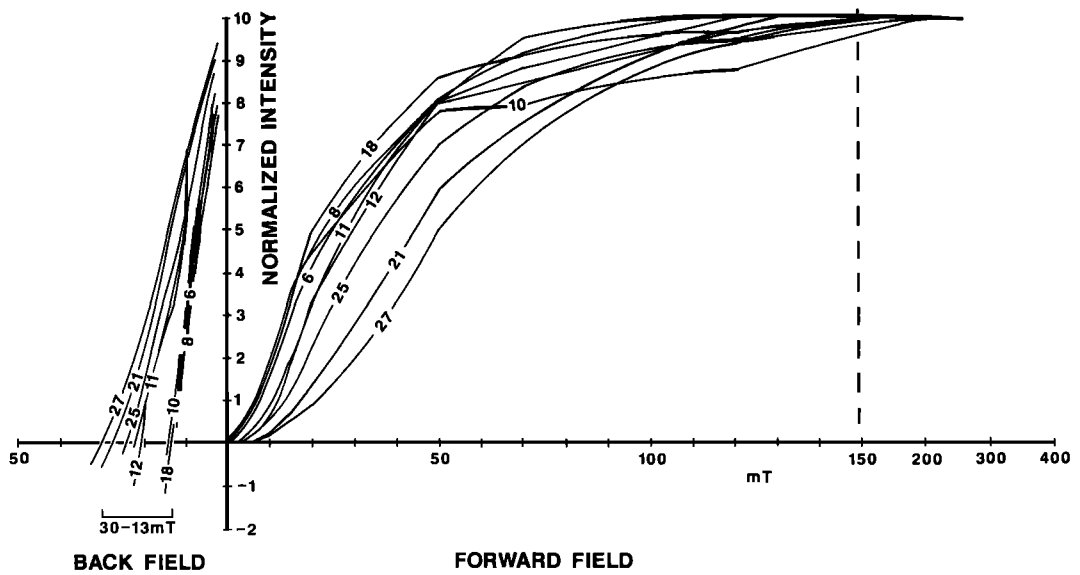


Fig. 7. Typical IRM acquisition curves for nine specimens from the Arkansas intrusions from nine representitive sites (labeled curves). Saturation IRM ranges from 150 to 200 mT, with a mean of 170 mT. Specimens have coercivities ranging from 15 to 30 mT. Average coercivities: Type A magnetization, 19 mT; type B magnetization, 17 mT; type C magnetization, 27 mT.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

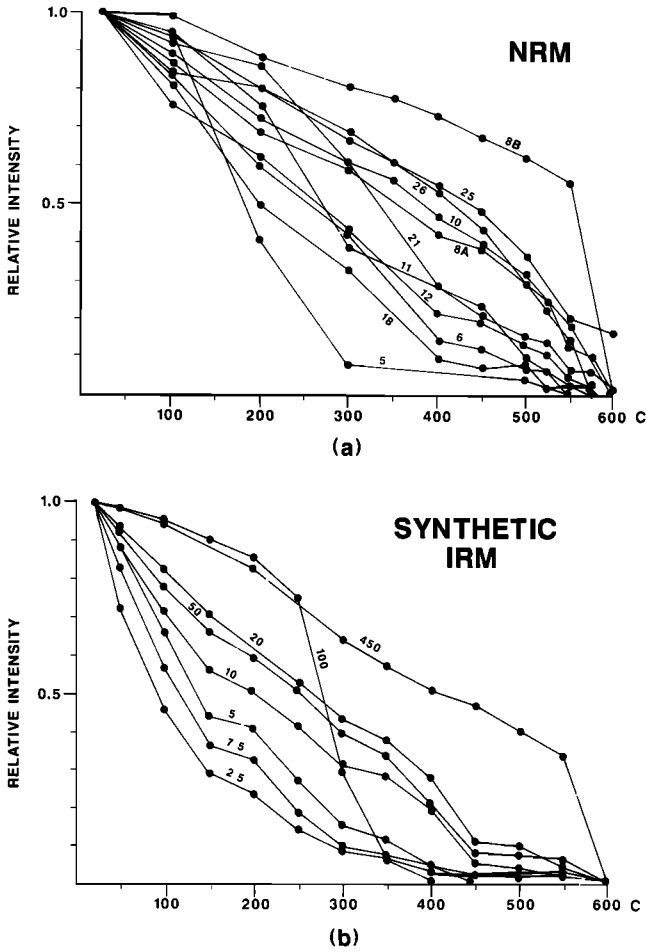


Fig. 8. (*a*) Changes in intensity during thermal demagnetization of type B and C magnetizations in 11 specimens from the Arkansas intrusions. Site numbers are shown above decay curves. (*b*) Changes in intensity during thermal demagnetization of isothermal remanent magnetization (IRM) produced in the laboratory; strengths of the dc fields, in milliteslas, are shown above each curve.

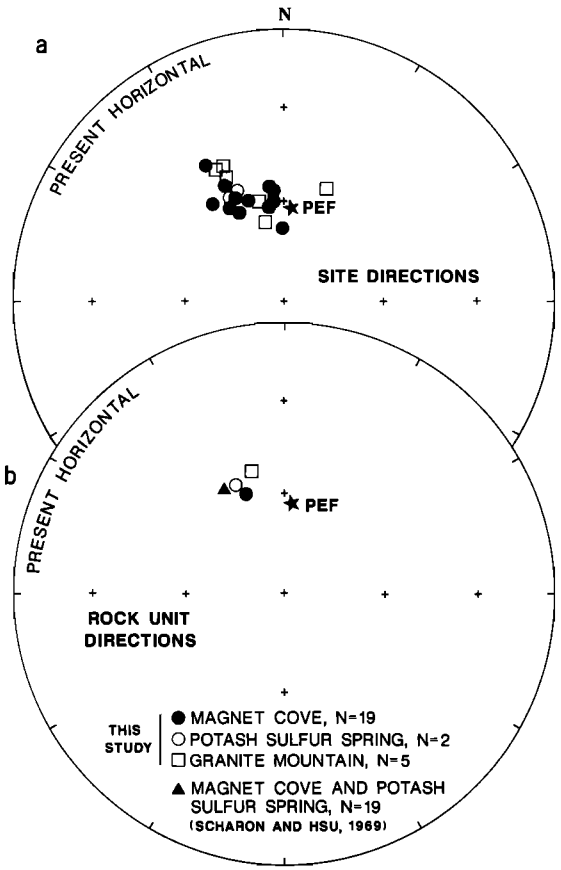


Fig. 9. Summary of the directions of magnetization for the Arkansas intrusions. (*a*) All site-mean directions with α_{95} less than 20°, and (*b*) locality-mean directions for individual intrusive complexes. *Scharon and Hsu*'s [1969] direction (triangle) is based on 19 MC/PS site means. Perimeter is the present-day horizontal.

9*b*, along with the results of *Scharon and Hsu* [1969]. All are significantly different from the present field. The mean directions determined in this study for MC/PS and GM (Table 3, intrusions 1 and 2) are not different ($P = 0.05$) by the method of *McFadden and Jones* [1981], despite an age difference of about 12 m.y. Furthermore, the overall mean directions determined in this and the previous study do not differ ($P = 0.05$), but the error is greatly reduced (compare Table 3, intrusions 3 and 4).

The paleopole for the middle Cretaceous (100–88 Ma) Arkansas intrusions, calculated from 20 site-mean VGPs (Table 2), is 193°E, 74°N, $K = 34$, and $A_{95} = 5.7°$. The corresponding paleolatitude for Little Rock, Arkansas (272.2°E, 34.4°N), is 37° ± 5°; using *Scharon and Hsu*'s [1969] paleopole it is 35° ± 8°.

GEOMAGNETIC PALEOSECULAR VARIATION DURING MID-CRETACEOUS TIME

Because the three igneous complexes sampled comprise several discrete cooling units intruded during a time span of about 12 m.y., the angular dispersion (S_d) of site-mean directions or site-mean VGPs should provide an estimate of the magnitude of the PSV. The angular dispersion is given by $81K^{-1/2}$ [*Watson and Irving*, 1957], where K is the precision of the site-mean VGPs. For the 20 site-mean VGPs the S_d is 13.6°, with upper (S_u) and lower (S_l) limits ($P = 0.05$) of 17.4° and 11.2°, respectively, using the method of *Cox* [1969]. This

TABLE 3. Locality-Mean Directions and Paleopoles for Alkalic Intrusions, Arkansas

Intrusion	*N*/*N*′	Direction				Paleopole			
		Dec, deg	Inc, deg	*k*	α_{95}	Longitude °E	Latitude °N	*K*	A_{95}
1, Magnet Cove, Potash Sulfur Springs	15/19	338.7	57.5	71.2	4.6	197.3	72.6	42.3	6.0
2, Granite Mountain	5/7	345.0	51.5	34.8	13.2	171.4	77.8	21.0	17.1
3, Intrusions 1 and 2 (this study)	20/26	340.4	56.0	56.8	4.4	192.5	74.1	34.0	5.7
4, Previous study	19/25	329.6	54.3	15.0	8.9	187.6	65.1	11.5	10.3

N/*N*′, number of sites used in determination of locality-mean directions and paleopoles, *N*′ total number of sites studied; other abbreviations same as for Table 2. Intrusion 4 is data of *Scharon and Hsu* [1969] for Magnet Cove and Potash Sulfur Springs.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4. Compilation of Mid-Cretaceous Paleopoles From the North American Craton

No.	Age, Ma	Rock Unit	Location	Paleopole Longitude °E	Paleopole Latitude °N	A_{95}	Plat	K
A	100 – 88	alkalic intrusions	Arkansas	193	74	5.7	37 ± 4	
B	129	lamprophyric dikes	Newfoundland	207	71	3.8	42 ± 3	
C*	136 – 118	alkalic intrusions	Quebec	191	73	3.2	37 ± 3	
D	109 ± 5	Isachsen diabase	N.W.T.	180	69	14.0	33 ± 11	
E†	85 – 89	Niobrara Formation	Kansas, Colorado, and Wyoming	192	66	6.0	37 ± 5	
1		A + B + C + D + E		192	71	4.3	37 ± 3	319
2		A + B + C + E – Preferred		196	71	4.9	38 ± 4	352
3		A + C + E		192	71	6.5	37 ± 5	344
4		A + C		192	73	3.8	39 ± 4	9930
5		A + B + C		197	73	4.7	37 ± 3	703
OP		overprint		201	68	9.4	···	31

Plat, mid-Cretaceous paleolatitude for Little Rock, Arkansas (272.2°E, 34.4°N); other abbreviations same as for Table 2. References: A, this study; B, *Prasad* [1981], *Lapointe* [1979]; C, *Foster and Symons* [1979]; D, *Larochelle and Black* [1963], *Larochelle et al.*, [1965]; E, *Shive and Frerichs* [1974].

**Eby* [1984] reported ages of 118 and 136 Ma for the alkalic intrusions of Quebec, although *Foland et al* [1986] report a more restricted age span of 124 plus or minus 1 Ma.

†Biostratigraphic age of Niobrara Formation (Coniacian-Santonian) converted to age in Ma, using time scale of *Harland et al.* [1982].

corresponds very well to the S_d (about 15°) predicted by a number of PSV models for a paleolatitude of 37° [*Cox*, 1962, 1970; *Irving and Ward*, 1964; *McElhinny and Merrill*, 1975; *Harrison*, 1980; *McFadden and McElhinny*, 1984]. Moreover, there is no evidence in our data that the PSV was particularly small during this period of infrequent polarity reversals.

Comparison With Other Cratonic North American Paleopoles

Other cratonic North American paleopoles of mid- to Late Cretaceous age are available from Newfoundland (B), Quebec (C), Northwest Territories (D), and Kansas-Wyoming-Colorado (E; Table 4 and Figure 10). Paleopoles for igneous

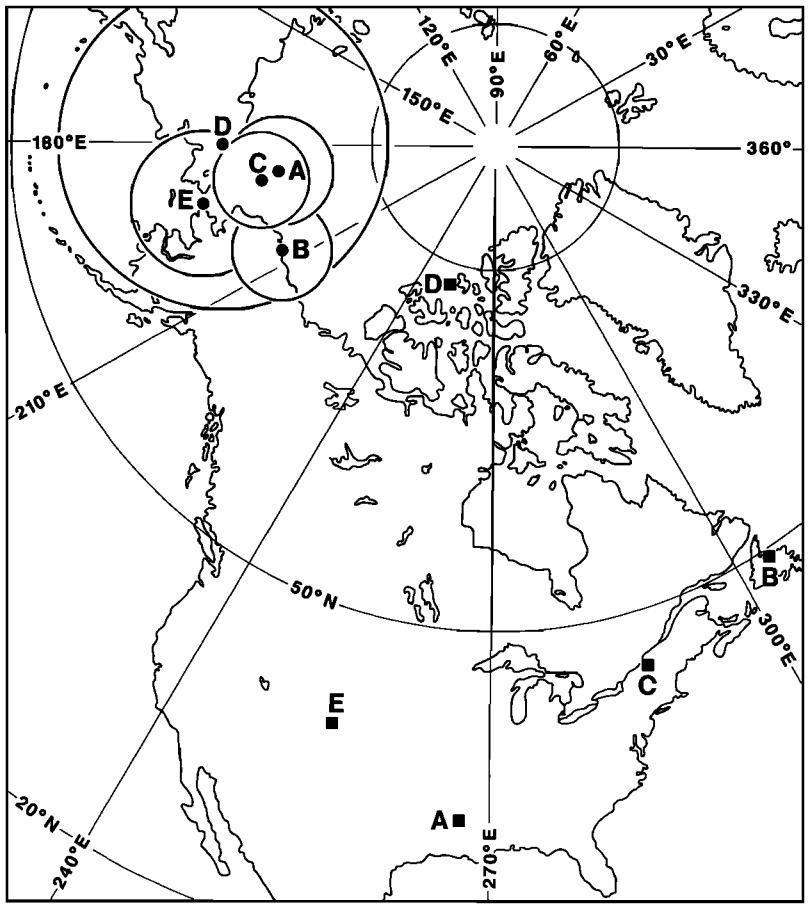


Fig. 10. Paleopole and confidence limits ($P = 0.05$) for the Arkansas intrusions (A) compared with four other middle Cretaceous paleopoles from cratonic North America: (B), lamprophyric dikes, Newfoundland; (C), igneous intrusions, Monteregian Hills, Quebec; (D), Isachsen diabasic rocks, Northwest Territories; (E), Niobrara Formation, Colorado, Kansas, and Wyoming. Squares show locations of study areas.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

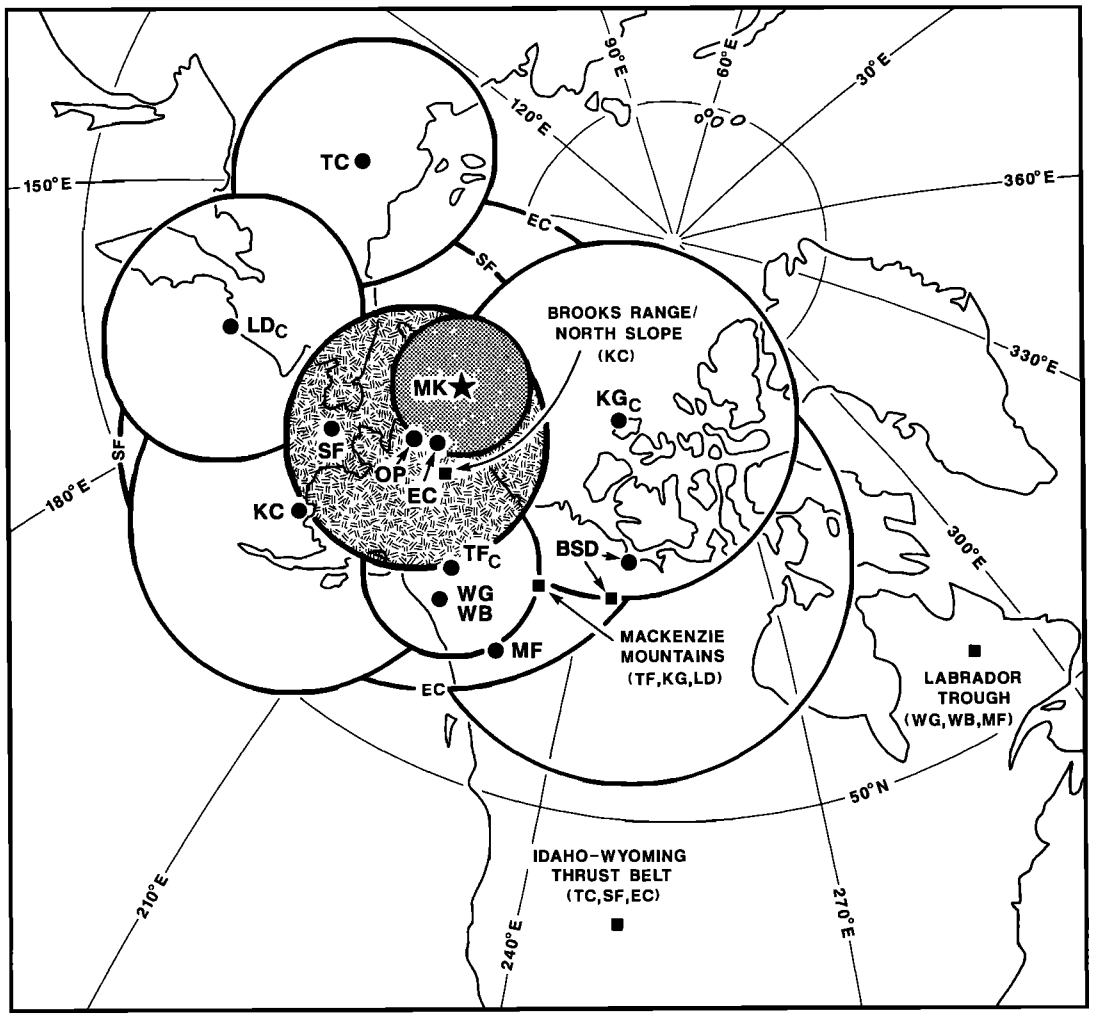


Fig. 11. Composite middle Cretaceous paleopole (MK) for cratonic North America (see Table 4, paleopole 2) compared with nine paleopoles representing possible mid- to Late Cretaceous remagnetizations of older rock units from the Mackenzie Mountains (Northwest Territories), Labrador Trough (Quebec), Brooks Range (Alaska), and the Wyoming-Idaho overthrust belt. Mackenzie Mountains, upper Proterozoic: TF_c, Tsezotene Formation, component C (219°E, 63°N, $A_{95} = 6°$) [*Park and Aitken*, 1986*a*]; KG_c, Katherine Group, component C (238°E, 77°N, $A_{95} = 11°$) [*Park and Aitken*, 1986*b*]; LD_c, Little Dal Group, component C (170°E, 60°N, $A_{95} = 9°$) [*Park*, 1981]. Labrador Trough, middle Proterozoic: WG/WB, Wakuach Gabbro and Willbob Basalt (238°E, 61°N, $A_{95} = 24°$); MF_c, Menihek Formation, component C (229°E, 59°N, $A_{95} = 26°$) [*Park*, 1977]. Brooks Range, Devonian-Mississippian: KC, Kanayut Conglomerate (197°E, 59°N, $A_{95} = 12°$) [*Hillhouse and Grommé*, 1983]. Wyoming-Idaho overthrust belt, Jura-Cretaceous: TC, Twin Creek Formation (145°E, 68°N, $A_{95} = 9°$) [McCabe et al., 1982]; SF, Stump Formation (170°E, 64°N, $A_{95} = 15°$), EC, Early Cretaceous sites (204°E, 69°N, $A_{95} = 17°$) [*Schwartz and van der Voo*, 1984]. Paleopoles derived from the "primary" magnetizations of those rock units for which mid- to Late Cretaceous remagnetizations are suggested: TF_m and LD_m; Tsezotene Formation, component A (146°E, 12°N), and Little Dal Group (141°E, 16°S); CT, Castignon Complex, Labrador Trough (268°E, 22°N) [*Park*, 1977]. The mean overprint paleopole, OP (201°E, 68°N, $A_{95} = 9.4°$), is determined with five of nine rock units assumed in situ; poles for LD_c, TC, SF, and EC are calculated with respect to bedding. OP is not significantly different ($P = 0.05$) from our preferred middle Cretaceous paleopole for cratonic North America, MK (196°E, 71°N, $A_{95} = 4.9°$).

intrusions in Arkansas (A) and the Monteregian Hills, Quebec (C) are virtually identical (Figure 10). This is significant because the Arkansas intrusions give ages between 88 and 100 Ma, whereas the Quebec intrusions, whose paleomagnetism is described by *Foster and Symons* [1979], range in age from 118 Ma to at least 124 Ma and perhaps to 136 Ma [*Eby*, 1984; *Foland et al.*, 1986]. The agreement of these two paleopoles indicates that there was little apparent polar wander relative to the North American craton for at least 30 m.y. (118–88 Ma) and perhaps as long as 48 m.y. (136–88 Ma).

The revised Arkansas paleopole is combined in several ways with previous data to provide a new mid-Cretaceous reference pole for cratonic North America (Table 4). The paleopole from the Isachsen diabase [*Larochelle and Black*, 1963; *Larochelle et al.*, 1965] has a large error ($A_{95} = 14°$) and is not included in our preferred combination (196°E, 71°N, $A_{95} = 4.9°$, $N = 4$). Because of the possibility of inclination error [e.g., *King*, 1955; *Griffiths et al.*, 1960], it might be considered desirable to use only data from igneous rocks, thereby omitting the paleopole from the Niobrara Formation. However, because

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the latter is not farsided (Figure 10, paleopole E), inclination errors are not present, so it is retained in our preferred combination.

CRATON AND CRATON-MARGIN OVERPRINTS

Studies of magnetic overprints observed in nine rock units from the craton and craton-margin ranging in age from Proterozoic to Jura-Cretaceous, have yielded directions with steep inclination and normal polarity. It is useful to compare their paleopoles with the newly derived cratonic reference pole (Figure 11). Numerous overprints of mid-Cretaceous age have been observed in allochthonous terranes of the western Cordillera [*Irving et al.*, 1985] but are not considered here. Five of nine paleopoles (from the Mackenzie Mountains (Northwest Territories), most from the Labrador Trough (Quebec), and Brooks Range (Alaska)) have been calculated with the rocks in situ. This is a reasonable approximation because tilt tests [*McFadden and Jones*, 1981] have been shown to be negative. The other four paleopoles, from the Labrador Trough (LD_c) and Idaho-Wyoming overthrust belt, have been calculated with the rocks restored to ancient horizontal.

Although overprinted paleopoles have large error circles, several (TC, LD_c, TF_c) still diverge significantly from the reference paleopole (Figure 11, MK). Note that MF, WG/WB, and LD_c diverge from MK in the direction of the "primary" magnetization of their respective rock units (see caption to Figure 11), and these divergences could be caused by incomplete separation of the overprint. The assumption that the overprint magnetization was acquired entirely with respect to present horizontal also may have caused some divergences. However, considered as a whole, the effects appear to be random because the mean "overprint paleopole" (OP of Table 4 and Figure 11) is not significantly different from the mid-Cretaceous cratonic paleopole.

The overprints are widespread and are carried by a variety of volcanic and sedimentary rocks. The phenomenon, therefore, may be comparable to the Kiaman (Late Carboniferous to Permian) overprinting which widely affects older rocks of the Applachian Fold Belt [*Irving and Opdyke*, 1965; *van der Voo et al.*, 1979; *Kent*, 1979] and the North American craton [*Irving and Strong*, 1985; *Miller and Kent*, 1986]. Both phenomena are associated with long intervals when reversals of the geomagnetic field did not occur. Both affect widely different rock types and must therefore involve a variety of processes [*Irving and Strong*, 1985; *Ellwood et al.*, 1986]. The overprinting episodes are associated in time with major diastrophism in the Cordillera during the mid-Cretaceous and with the Alleghenian orogeny in the Appalachian Fold Belt during Permo-Carboniferous time. Continent-wide migration of fluids during orogenesis [*Leach et al.*, 1984; *Oliver*, 1986, 1987], is clearly a potent mechanism for producing overprints of low-temperature chemical origin. Support for this mechanism comes from $^{40}Ar/^{39}Ar$ dating of authigenic potassium feldspar in lower Paleozoic limestones from the Valley and Ridge Province of the Appalachian orogen, which indicates that these detrital feldspars are Late Carboniferous to Early Permian in age [*Hearn et al.*, 1987]. They suggest that the alteration did not occur isochemically but required the influx of pore fluids through the rocks, reflecting regional fluid migration events during the Late Paleozoic Alleghenian orogeny. Clearly, such a process could produce new (secondary) iron minerals as well. If the hypothesis of remagnetization by periodic crustal flushing is correct, then it is evident that accurate correlation of overprints with well-established cratonic reference poles provides a very powerful means of assisting in mapping and dating its occurrence.

The evidence of Figure 11 shows that generally such accurate correlations cannot at present be made because errors in determination of overprint directions are large. A critical test of the hypothesis of a continent-wide remagnetization episode during the mid-Cretaceous would be to determine whether these individual divergences are real, or artefacts of unrecognized technical complexities in data from the craton and craton-margin, as we propose above.

Acknowledgments. The work was supported by U.S. National Science Foundation grant EAR-8417639 to R. S. Coe, to whom we are grateful for support, and by a Natural Sciences and Engineering Research Council of Canada Post-Doctoral Fellowship to B. R. Globerman. We are very grateful to J. M. Howard (Arkansas Geological Commission), who provided geologic guidance in the Magnet Cove area and made the necessary arrangements for sampling on private lands. We are also grateful to B. Petty for helpful assistance in the field and to R. Burcham (Granite Mountain quarries), H. deLinde (Diamond Jo quarry), J. Lowrey (3M Company), and D. Owens (UMETCO Minerals) for permission to sample on their properties. We would also like to thank E. M. Morris and N. Eby for their advice on the petrology and age relations of the Granite Mountain stock. Finally, we thank G. N. Eby, H. Gabrielse, J. M. Howard, E. Mankinen, and two anonymous reviewers for their helpful comments. Geological Survey of Canada contribution 15588.

REFERENCES

Cox, A. V., The remanent magnetization of some Cenozoic volcanic rocks, Ph.D. dissertation, 145 pp., Univ. of Calif., Berkeley, 1959.

Cox, A., Anomalous remanent magnetization of basalt, *U.S. Geol. Surv. Bull., 1083E*, 131–160, 1961.

Cox, A., Analysis of present geomagnetic field for comparison with paleomagnetic results, *J. Geomagn. Geoelectr., 13*, 101–112, 1962.

Cox, A., Angular dispersion due to random magnetization, *Geophys. J. R. Astron. Soc., 8*, 345–355, 1964.

Cox, A., Confidence limits for the precision parameters *k*, *Geophys. J. R. Astron. Soc., 18*, 545–549, 1969.

Cox, A., Latitude dependence of the angular dispersion of the geomagnetic field, *Geophys. J. R. Astron. Soc., 20*, 253–269, 1970.

Creer, K. M., A preliminary palaeomagnetic survey of certain rocks in England and Wales, Ph.D. dissertation, 203 pp., Queens Coll., Cambridge, England, 1955.

Creer, K. M., An analysis of the geomagnetic field using palaeomagnetic methods, *J. Geomagn. Geoelectr., 13*, 113–119, 1962.

Creer, K. M., E. Irving, and A. E. M. Nairn, Palaeomagnetism of the Great Whin Sill, *Geophys. J. R. Astron. Soc., 2*, 306–323, 1959.

Eby, G. N., Geochronology of the Monteregian Hills alkaline igneous province, Quebec, *Geology, 12*, 468–470, 1984.

Eby, G. N., Fission-track geochronology of the Arkansas alkaline province, *U.S. Geol. Surv. Open File Rep., 87-0287*, 14 pp., 1987.

Ellwood, B. B., W. Balsam, B. Burkart, G. J. Long, and M. L. Buhl, Anomalous magnetic properties in rocks containing the mineral siderite: Paleomagnetic implications, *J. Geophys. Res., 91*, 12,779–12,790, 1986.

Erickson, R. L., and L. V. Blade, Map of bedrock geology of Magnet Cove igneous area, Hot Spring County, Arkansas, scale 1:6000, *U.S. Geol Surv. Misc. Invest. Field Stud. Map, MF-53*, 1956.

Erickson, R. L., and L. V. Blade, Geochemistry and petrology of the alkalic igneous complex at Magnet Cove, Arkansas, *U.S. Geol. Surv. Prof. Pap., 425*, 95 pp., 1963.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A, 217*, 295–305, 1953.

Foland, K. A., L. A. Gilbert, C. A. Sebring, and J.-F. Feng, $^{40}Ar/^{39}Ar$ ages for plutons of the Monteregian Hills, Quebec: Evidence for a single episode of Cretaceous magmatism, *Geol. Soc. Am. Bull., 97*, 966–974, 1986.

Foster, J., and D. T. A. Symons, Defining a paleomagnetic polarity

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

pattern in the Monteregian intrusives, *Can. J. Earth Sci., 16*, 1716–1725, 1979.

Frei, L. S., J. R. Magill, and A. Cox, Paleomagnetic results from the central Sierra Nevada: Constraints on reconstruction of the western United States, *Tectonics, 3*, 157–177, 1984.

Gordon, R. G., A. Cox, and S. O'Hare, Paleomagnetic Euler poles and the apparent polar wander and absolute motion of North America since the Carboniferous, *Tectonics, 3*, 499–537, 1984.

Graham, K. W. T., The remagnetization of a surface outcrop by lightning currents, *Geophys. J. R. Astron. Soc., 6*, 85–102, 1961.

Griffiths, D. H., R. F. King, A. I. Rees, and A. E. Wright, Remanent magnetism of some recent varved sediments, *Proc. R. Soc. London, Ser. A, 256*, 359–383, 1960

Grommé, C. S., and R. T. Merrill, Paleomagnetism of Late Cretaceous plutons in the Sierra Nevada, California: Further results, *J. Geophys. Res., 70*, 3407–3420, 1965.

Grommé, C. S., R. T. Merrill, and J. Verhoogen, Paleomagnetism of Jurassic and Cretaceous plutonic rocks in the Sierra Nevada and its significance for polar wandering and continental drift, *J. Geophys. Res., 72*, 5661–5684, 1967.

Haley, B. R., E. E. Glick, W. V. Bush, B. F. Clardky, C. G. Stone, M. B. Woodward, and D. L. Zachry, Geologic map of Arkansas, scale 1:500,000, U.S. Geol. Surv., Reston, Va., 1976.

Harland, W. B., A. V. Cox, P. G. Llewellyn, C. A. G. Pickton, A. G. Smith, and R. Walters, *A Geologic Time Scale*, 131 pp., Cambridge University Press, New York, 1982.

Harrison, C. G. A., Secular variation and excursions of the Earth's magnetic field, *J. Geophys. Res., 85*, 3511–3522, 1980.

Hearn, P. P., J. F. Sutter, and H. E. Belkin, Evidence for late Paleozoic brine migration in Cambrian carbonate rocks of the central and southern Appalachians: Implications for Mississipi Valley-type sulfide mineralization, *Geochim. Cosmochim. Acta, 51*, 1323–1334, 1987.

Hillhouse, J. W., and C. S. Grommé, Paleomagnetic studies and the hypothetical rotation of Arctic Alaska, *J. Alaska Geol. Soc., 2*, 27–39, 1983.

Howard, J. M, Transition element geochemistry and petrography of the Potash Sulfur Springs intrusive complex, Garland County, Arkansas, M. S. thesis, 118 pp., Univ. of Arkansas, Fayetteville, 1974.

Howard, J. M., Geology of the Wilson Springs (Potash Sulfur Springs) vanadium deposit, Garland County, Arkansas, in *Sedimentary and igneous rocks of the Ouachita Mountains of Arkansas, Guideb. 86.2*, edited by C. G. Stone, J. M. Howard, and B. R. Haley, pp. 99-106, Arkansas Geological Commission, Little Rock, 1986.

Humboldt, A. von, Uber die merkwürdige magnetische Polarität einer Gebirgskuppe von Serpentinstein, *Greus Neues J. Phys., 4*, 136–140, 1797.

Irving, E., *Paleomagnetism and Its Application to Geological and Geophysical Problems*, 399 pp., John Wiley, New York, 1964.

Irving, E., Paleopoles and paleolatitudes of North America and speculations about displaced terrains, *Can. J. Earth Sci., 16*, 669–694, 1979.

Irving, E, and N. D. Opdyke, The palaeomagnetism of the Bloomsburg red beds and its possible application to the tectonic history of the Appalachians, *Geophys. J. R. Astron. Soc, 9*, 153–167, 1965.

Irving, E, and D. F. Strong, Paleomagnetism of rocks from Burin Peninsula, Newfoundland: Hypothesis of late Paleozoic displacement of Acadia criticized, *J. Geophys. Res., 90*, 1949–1962, 1985.

Irving, E., and M. A. Ward, A statistical model of the geomagnetic field, *Pure. Appl. Geophys., 57*, 47–52, 1964.

Irving, E., G. J. Woodsworth, and P. J Wynne, Paleomagnetic evidence for displacement from the south of the Coast Plutonic Complex, British Columbia, *Can. J Earth Sci., 22*, 584–598, 1985.

Kent, D. V., Paleomagnetism of the Devonian Onondaga Limestone revisited, *J Geophys. Res., 84*, 3576–3588, 1979.

Kent, J. T., J. C Briden, and K. V. Mardia, Linear and planar structure in ordered multivariate data, as applied to progressive demagnetization of paleomagnetic remanence, *Geophys. J. R. Astron. Soc., 75*, 593–621, 1983.

King, R F., Remanent magnetism of artificially deposited sediments, *Mon. Not. R. Astron. Soc., Geophys. Supp., 7*, 115–123, 1955.

Kobayashi, K., Chemical remanent magnetization of ferromagnetic minerals and its application to rock magnetism, *J. Geomagn. Geoelectr., 10*, 99–117, 1959.

Lapointe, P. L., Paleomagnetism of the Notre Dame lamprophyre dikes, Newfoundland, and the opening of the North Atlantic Ocean, *Can. J. Earth Sci., 16*, 1823–1831, 1979.

Larochelle, A., Application of paleomagnetism to geological correlation, *Nature, 192*, 37, 1961.

Larochelle, A. Paleomagnetism of the Monteregian Hills, southeastern Quebec, *Geol. Surv. Can. Bull., 79*, 1043, 1962.

Larochelle, A., and R. R. Black, An application of paleomagnetism in estimating the age of rocks, *Nature, 198*, 1260–1262, 1963.

Larochelle, A., R. F. Black, and R. K. Wanless, Paleomagnetism of the Isachsen diabasic rocks, *Nature, 208*, 179, 1965.

Leach, D. L., J. G. Viets, and L. Rowan, Appalachian-Ouachita orogeny and Mississippi Valley-type lead-zinc deposits, *Geol. Soc. Am. Abstr. Programs, 16*, 572, 1984.

Mankinen, E. A., Paleomagnetic evidence for a Late Cretaceous deformation of the Great Valley sequence, Sacramento Valley, California, *J. Res. U.S. Geol. Surv., 6*, 383–390, 1978.

McCabe, C., R. van der Voo, and B. H. Wilkinson, Paleomagnetic and rock magnetic results from the Twin Creek Formation (Middle Jurassic), Wyoming, *Earth Planet. Sci. Lett., 60*, 140–146, 1982.

McCormick, G. R., Ferromagnetism mineral chemistry of Arkansas carbonatite intrusions, *Geol. Assoc. Can. Programs Abstr., 11*, 100, 1986.

McElhinny, M. W., *Palaeomagnetism and Plate Tectonics*, 358 pp., Cambridge University Press, New York, 1973.

McElhinny, M. W., and R. T. Merrill, Geomagnetic secular variation over the past 5 m.y., *Rev. Geophys., 13*, 687–708, 1975.

McFadden, P. L., and D. L. Jones, The fold test in palaeomagnetism, *Geophys. J. R. Astron. Soc., 67*, 53–58, 1981.

McFadden, P. L., and M. W. McElhinny, A physical model for palaeosecular variation, *Geophys. J. R. Astron. Soc., 78*, 809–830, 1984.

Miller, J., and D. V. Kent, Synfolding and prefolding magnetizations in the Upper Devonian Catskill Formation of eastern Pennsylvania: Implications for the tectonic history of Acadia, *J. Geophys. Res., 91*, 12,791–12,803, 1986.

Morris, E. M., The Cretaceous Arkansas alkalic province: A summary of petrology and chemistry, *Spec. Pap. Geol. Soc. Am., 215*, 216–233, 1987.

Morris, E. M., and G. N. Eby, Petrologic and age relations in Granite Mountain syenite, *Geol. Soc. Am. Abstr. Programs, 17*, 198, 1986.

Naeser, C. W., and H. Faul, Fission track annealing in apatite and sphene, *J. Geophys. Res., 74*, 705–710, 1969.

Oliver, J., Fluids expelled tectonically from orogenic belts: Their role in hydrocarbon migration and other geologic phenomena, *Geology, 14*, 99–102, 1986.

Oliver, J., Synthesis of observations related to brines expelled tectonically from orogenic belts, *Eos Trans. AGU, 68*, 291, 1987.

Park, J. K., A reconnaissance paleomagnetic survey of the central Labrador Trough, Quebec, *Can. J. Earth Sci., 14*, 159–174, 1977.

Park, J. K., Analysis of multicomponent magnetism of the Little Dal Group, Mackenzie Mountains, Northwest Territories, *J. Geophys. Res., 86*, 5134–5146, 1981.

Park, J. K., and J. D. Aitken, Paleomagnetism of the late Proterozoic Tsezotene Formation of the Mackenzie Mountains, northwestern Canada, *J. Geophys. Res., 91*, 4955–4970, 1986*a*.

Park, J. K., and J. D. Aitken, Paleomagnetism of the Katherine Group in the Mackenzie Mountains: Implications for post-Grenville (Hadrynian) apparent polar wander, *Can. J. Earth Sci., 23*, 308–323, 1986*b*.

Petrova, G. N., Magnetic stability of rocks, *Izv. Akad. Nauk, 1*, 57–67, 1957.

Prasad, J. N., Paleomagnetism of the Mesozoic lamprophyre dikes in north-central Newfoundland, M S. thesis, 119 pp., Mem. Univ. of Newfoundland, St. John's 1981.

Runcorn, S. K., Paleomagnetic survey in Utah and Arizona: Preliminary results, *Geol Soc Am. Bull., 67*, 301–316, 1956.

Scharon, L., and I-C Hsu, Paleomagnetic investigation of some Arkansas alkalic igneous rocks, *J Geophys. Res., 74*, 2774–2779, 1969.

Schwartz, S. Y., and R. van der Voo, Paleomagnetic study of thrust sheet rotation during foreland impingement in the Wyoming-Idaho overthrust belt, *J. Geophys. Res., 89*, 10,077–10,086, 1984.

Shive, P N., and W E. Frerichs, Paleomagnetism of the Niobrara Formation in Wyoming, Colorado, and Kansas, *J. Geophys. Res., 79*, 3001–3007, 1974.

van der Voo, R., A N. French, and R. B. French, A paleomagnetic pole position from the Upper Devonian Catskill red beds and its tectonic implications, *Geology, 7*, 345–348, 1979.

Watson, G. S. and E. Irving, Statistical methods in rock magnetism, *Mon. Not. R. Astron. Soc., 7*, 289–300, 1957.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Williams, J. F., The igneous rocks of Arkansas, *Arkansas Geol. Surv. Annu. Rep.*, *2*, 457 pp., 1891.

Zartman, R. E. Geochronology of some alkalic rock provinces in eastern and central United States, *Annu. Rev. Earth Planet. Sci.*, *5*, 257–286, 1977.

Zartman, R. E., and J. M. Howard, Uranium-lead age of large zircon crystals from the Potash Sulfur Springs igneous complex, Garland County, Arkansas, *Spec. Pap. Geol. Soc. Am.*, *215*, 235–240, 1987.

Zartman, R. E., M. R. Brock, A. V. Hey, and H. H. Thomas, K-Ar and Rb-Sr ages of some alkalic intrusive rocks from central and eastern United States, *Am. J. Sci.*, *265*, 848–870, 1967.

B. R. Globerman and E. Irving, Pacific Geoscience Centre, Geological Survey of Canada, Box 6000, Sidney, B.C., Canada V8L 4B2.

(Received August 21, 1987;
revised March 25, 1988;
accepted March 28, 1988.)

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11721 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License